HARRISON PIELKE-LOMBARDO

Full-Stack Software Engineer & Computational Biologist

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Full-stack Software Engineer and Computational Biologist with 8+ years of experience building production-grade AI systems. Expert in bridging the gap between complex biomedical knowledge graphs and modern AI applications. Proven track record of driving $2.1M in annual revenue through deployment of LLM-based automation and high-volume data pipelines. Deep experience with clinical ontologies and human-in-the-loop annotation workflows.

TECHNICAL SKILLS

AI/ML & GenAI: LLMs, PyTorch, NLP, Claude, Copilot, Statistical Inference
Knowledge Engineering: Knowledge Graphs, Semantic Web (RDF), Ontologies (Protégé), Causal Reasoning, ICD, SNOMED
Languages & Frameworks: Python, Clojure/ClojureScript, Java, JavaScript, SQL, GraphQL, Bash
Cloud & Infrastructure: AWS (Lambda, EC2, ECS, SQS, SNS, RDS, DynamoDB, CloudFront), Terraform, Docker, PostgreSQL, Datomic, Redis, Git/GitHub
Healthcare AI & MedTech: Clinical NLP, Automated Relation Extraction, Explainable AI (XAI), Human-in-the-Loop Annotation

EXPERIENCE

Senior Software Engineer
Sep 2022 - Present
Peruse Technology Inc (SOLD)
Remote

- Scaled product from zero to $2.1M ARR and 70+ B2B clients, leading to a successful acquisition.
- Built a custom rules engine and core load processing pipeline using Clojure and Python (REST / GraphQL).
- Trained and deployed LLM-based ML models using PyTorch to automate logistics document processing and broker verification.
- Managed cloud infrastructure via Terraform on AWS (Lambda , ECS , SQS , RDS) to support high-volume document workflows.
- Executed a complex database migration from Datomic to Postgres while maintaining 100% system uptime.

Computational Biologist
Aug 2016 – Aug 2022
University of Colorado, Anschutz Medical Campus
Aurora, CO

- Managed interconnected computational projects across 4+ cross-functional teams, producing 6 publications, 4 dataset releases, and 5 package releases.
- Harmonized large-scale biomedical knowledge graphs represented in RDF , Datomic , and SQL formats for drug-target identification and diagnostic assistance applications.
- Implemented a novel bootstrapping approach for automated relation extraction; optimized with GPU-accelerated matrix operations to improve drug-target precision by 10%.
- Developed algorithmic approaches in Clojure and Python for extending analogical reasoning of biological networks, enabling human-interpretable pattern discovery.

SELECTED PROJECTS

Knowtator: Semantic Annotation & Ontology Platform
2016–2022

- Deployed a Java / ClojureScript web application for ontology development and text annotation with interactive graph visualization.
- Facilitated creation of 100k+ concept and relation annotations using ICD and SNOMED terminologies, evolving the system through iterative feedback from clinicians and researchers.
- Implemented CI/CD pipelines to support human-in-the-loop annotation workflows for validating LLM outputs against medical data models.

PRESENTATIONS

IDEAL Invited Speaker

June 2024

The Institute for Data, Econometrics, Algorithms, and Learning

Chicago, IL

- Title: AI in the Transportation Sector: Challenges and Successes

EDUCATION

Master of Science in Computational Bioscience

2022

University of Colorado, Anschutz Medical Campus

Aurora, CO

Bachelor of Science in Applied Mathematics

2016

University of Colorado

Boulder, CO

- Minors: Computer Science & Biochemistry